PH 5110: Optics and Photonics

Term Paper Abstract Submission

TOP-Q: Can Topological Edge Modes Protect Interferometric Phase? Implications for Neuromorphic and Quantum Photonic Circuits

Kotapati Kishore (Roll No. EE26E016)

Abstract

Programmable photonic processors realise unitary transformations through meshes of Mach–Zehnder interferometers (MZIs), supporting both optical neural networks and multiphoton quantum interference. Their accuracy is limited by fabrication-induced variations in waveguide width and gap, which perturb propagation constants and couplings, introducing phase and splitting-ratio errors that compound across the mesh. Topological photonics offers symmetry-based robustness, but this protection holds only for perturbations that preserve the relevant symmetry and keep the band gap open; it does not suppress loss or symmetry-breaking detuning. This term paper, framed as TOP-Q (Topologically Protected Optical Processing for Quantum and Neuromorphic Photonics), addresses one question: does topological protection of optical transport also protect the interferometric phase on which photonic processing depends? Using coupled-mode theory, a Su–Schrieffer–Heeger coupled-waveguide array will be characterised by its Bloch dispersion, winding number, bulk–boundary correspondence and edge-mode localisation length, and compared with a trivial-defect array. Monte Carlo disorder ensembles will contrast coupling disorder, which preserves chiral symmetry and is expected to pin the edge-mode propagation constant, with on-site detuning, which breaks it; propagation loss will be treated as a separate mechanism. These phase statistics will be propagated through MZI and mesh transfer matrices to evaluate output-intensity error and a normalised unitary fidelity. As a quantum extension, residual splitting-ratio imbalance and modal mismatch will be mapped onto Hong–Ou–Mandel visibility, distinguishing photon distinguishability from decoherence. The study aims to identify which error channels topological design can suppress and which it cannot, yielding a physically bounded design criterion for robust photonic circuits.

Keywords: topological photonics; Su–Schrieffer–Heeger model; chiral symmetry; Mach–Zehnder interferometer meshes; photonic neural networks; Hong–Ou–Mandel interference

Annotated References

[1] M. Reck, A. Zeilinger, H. J. Bernstein, and P. Bertani, *Phys. Rev. Lett.* **73**, 58 (1994). *Basis of the MZI matrix model.*

[2] Y. Shen *et al.*, *Nat. Photonics* **11**, 441 (2017). *Neuromorphic application whose accuracy depends on phase control.*

[3] M. Y.-S. Fang *et al.*, *Opt. Express* **27**, 14009 (2019). *Imprecision in optical neural networks.*

[4] T. Ozawa *et al.*, *Rev. Mod. Phys.* **91**, 015006 (2019). *Symmetry conditions and limits of protection.*

[5] W. P. Su, J. R. Schrieffer, and A. J. Heeger, *Phys. Rev. Lett.* **42**, 1698 (1979). *Original model and edge states.*

[6] A. Blanco-Redondo *et al.*, *Phys. Rev. Lett.* **116**, 163901 (2016). *Topological versus trivial modes in silicon.*

[7] C. K. Hong, Z. Y. Ou, and L. Mandel, *Phys. Rev. Lett.* **59**, 2044 (1987). *Defines the coincidence dip and visibility.*

[8] A. Blanco-Redondo *et al.*, *Science* **362**, 568 (2018). *Biphoton states robust to disorder.*